Toward Efficient Remote Sensing Image Change Detection via Cross-Temporal Context Learning

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***Abstract*— Change detection (CD) aims to find areas of specific changes in multitemporal remote sensing images. The existing methods fail to adequately explore the cross-temporal global context, making the establishment of spatial–temporal deep global associations insufficient and inefficient. As a result, their performance is vulnerable to complex and various objects in changing scenes. Hence, we propose a cross-temporal context learning network, termed as CCLNet, where the intratemporal and intertemporal long-range dependencies are mined and interactively fused, to fully exploit the cross-temporal context information. Specifically, a lightweight convolutional neural network (CNN) is first used to extract deep semantic features. Then, a well-designed cross-temporal fusion transformer (CFT) is proposed to locate the changing objects in the scene by establishing the long-range dependence across bitemporal images. Thanks to this, temporal-specific information extraction and cross-temporal information integration are seamlessly integrated into the same network, thereby significantly improving the discriminative features of changing objects. Furthermore, this allows us to use naive backbones with low computational costs to achieve reliable CD performance. Experiments on mainstream benchmarks show that our proposed method can handle CD tasks faster than state-of-the-art (SOTA) methods while maintaining better or comparable matching accuracy on a single RTX3090.**

***Index Terms*— Attention mechanism, change detection (CD), high-resolution optical remote sensing image.**

I. Introduction

CHANGE detection (CD) aims to detect relevant changes in registered remote sensing images from different times [1]. Such change usually has different definitions according to different applications. An in-depth study of CD tasks is of great value for many applications. For urban expansion [2], [3], fine details of architectural changes are helpful for urban management. For damage assessment [4], [5], accurate identification of buildings damaged by natural disasters is important for quantifying damage and arranging rescue. For vegetation change [6], the growth trend of vegetation is an important part of ecological monitoring. In general, detecting relevant changes of interest from complex scenes is a challenging task due to the following two reasons. First, imaging conditions such as atmosphere, season, and illumination at different times will cause changes in certain areas of the image. Second, the complexity of the scene makes it difficult to detect some changing objects. For example, the object may have similar spatial and spectral features to the background.

Due to the heavy reliance on manual operations, traditional CD methods are difficult to achieve satisfactory performance [7], [8], [9], [10], [11]. With the rapid development of deep learning in the computer vision (CV) field in recent years, deep models continue to refresh the performance of CD task. Nowadays, there are two main categories of CD methods with excellent performance. They are methods based on convolutional neural networks (CNNs) [12], [13], [14], [15], [16], [17], [18], [19] and transformer networks [20], [21], respectively. The CNN-based methods usually enhance the representation ability of features by stacking more convolutional layers and utilizing various spatial or channel attention modules. However, due to the inherent receptive field of convolution, they are struggling to establish long-range associations in space and time. Since changes of interest usually require sufficient context information within the spatial and temporal scope to be detected in complex scenes, the modeling of global information has become the top priority to further improve the CD task. The transformer networks with strong long-range dependence modeling capabilities are soon introduced in the CD community and get competitive performance. Nonetheless, existing transformer-based methods mainly focus on intratemporal global context mining rather than intertemporal, which leads to an insufficient exploration of spatial–temporal deep global context, hindering the further improvement of CD performance.

To solve the above problem, we propose a cross-temporal context learning network (CCLNet), which achieves efficient CD via intertemporal and intratemporal global modeling, as shown in Fig. 1. The design of CCLNet is mainly considered the following points. From the perspective of overall

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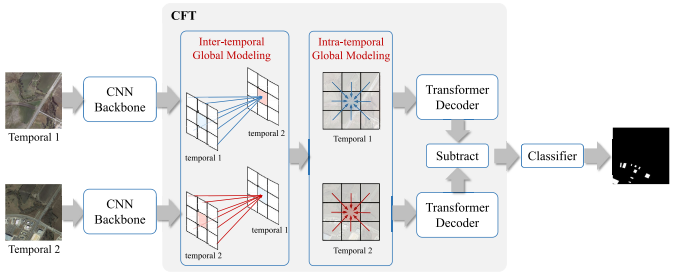


Fig. 1. Illustration of CCLNet. For the input bitemporal images, CNN is first used to extract deep semantic features. Then, the proposed cross-temporal fusion transformer (CFT) is used to fully mine the cross-temporal global context by intertemporal and intratemporal long-range dependence modeling, so as to improve the discriminative of object features. Finally, the features generated by the CFT are fed into the classifier head to obtain the change prediction.

network design, the superior inductive bias of convolution makes it very suitable for initial feature extraction [22]. Also, using convolution for shallow feature modeling can be slightly faster than transformer with the computational complexity $O(n^2)$. By combining shallow modeling of CNN and deep modeling of the transformer, shallow spatial and deep semantic information can be fully mined, laying the foundation for subsequent cross-temporal feature fusion. From the perspective of cross-temporal context learning, we first establish global interactions of bitemporal features, and then aggregate global information in different temporal features. Through this sequence of intertemporal and intratemporal information extraction, cross-temporal and temporal-specific information integration are seamlessly integrated, and thus, cross-temporal contextual information is maximally mined.

Specifically, a naive CNN is first used for local modeling to extract deep semantic features. Then, a novel CFT is proposed to establish cross-temporal deep global associations by sequentially performing intertemporal and intratemporal long-distance dependence modeling, thus boosting feature discriminativeness of changing objects. In CFT, the input bitemporal deep features are converted into tokens by the overlapping patch embedding strategy. Multilayer perceptron (MLP) is then utilized to perform deep semantic associations (DSAs) across all tokens in both spatial and channel dimensions. Next, a transformer-based cross-temporal encoder models long-range dependencies on intertoken and intratoken to learn cross-temporal deep context. The transformer decoder (TD) is finally used to map deep semantic features into the desired space from a global perspective, which greatly preserves the cross-temporal global context information in deep semantics compared with other decoding operations. After CFT, the bitemporal feature maps generated by the decoder are sent to the classifier to obtain the final detection map. In summary, our contributions are as follows.

1) We propose an efficient framework for CD of remote sensing images. The proposed framework can fully utilize local and global information for better complementary feature integration, thus achieving efficient performance.
2) A well-designed CFT is proposed to achieve efficient extraction of cross-temporal global context information by effectively mining and fusing intertemporal and intratemporal long-range dependencies.
3) Experiments on three benchmarks LEarning, Vision, and Remote sensing (LEVIR-CD) [12], deeply supervised image fusion network (DSIFN-CD) [23], and WuHan University (WHU-CD) [24] demonstrate the superiority of our framework, which confirms the positive effects of cross-temporal context learning.

II. Related Work

A. Deep-Learning-Based RS Image CD

As a key technology in remote sensing image processing, CD continues to attract research interest in recent years. The mainstream deep-learning-based supervised CD methods can be divided into three categories according to the fusion level: image level, feature level, and decision level.

The image-level methods [25], [26], [27], [28], [29], [30] first concatenate the bitemporal images into a six-channel image and then send it to the neural network to obtain the CD prediction. The feature-level methods send the bitemporal images to the network, respectively, and directly obtain the change prediction through feature fusion decoding. Among the feature-level methods, some methods [31], [32], [33] first split the bitemporal image into pairs of patches, and a CNN is applied to each pair of patches to obtain the detection of its center. This kind of method regards the CD task as a similarity detection process. Other methods belonging to the feature-level fuse the bitemporal image features extracted by the backbone, and directly decode pixel by pixel to obtain predictions [12], [13], [14], [15], [16], [17], [18], [19]. The decision-level methods [34], [35], [36] perform semantic segmentation on each image individually. Two segmentation results are then compared to generate change results. This kind of method is only practical when there are additional semantic labels for bitemporal images.

Most of the recent work falls into the category of feature-level and enhances feature-discriminative representations to improve performance by multiscale feature fusion, spatial–temporal context modeling, or combining generative adversarial network (GAN)-based optimization objectives. Hou et al. [18] proposed an end-to-end dual-branch architecture W-Net, applying the recently popular GAN by reformulating CD as an image translation problem. Chen and Shi [12] designed a CD self-attention module, which generates more discriminative features by modeling the spatial–temporal relationships between any two pixels at different times and positions. At the same time, considering that objects may have different scales, it divides the image into multiscale subregions and uses the designed module in each subregion. Liu et al. [15] proposed a dual-task constrained deep Siamese convolutional network (DTCDSCN), which makes features more object-level discriminative by performing both CD and semantic segmentation tasks. Furthermore, a dual attention module (DAM) is proposed to model the contextual relationship between channels and spatial positions. Fang et al. [37] proposed a densely connected Siamese network SNUNet-CD for CD, which improves the combination of deep semantic and shallow spatial

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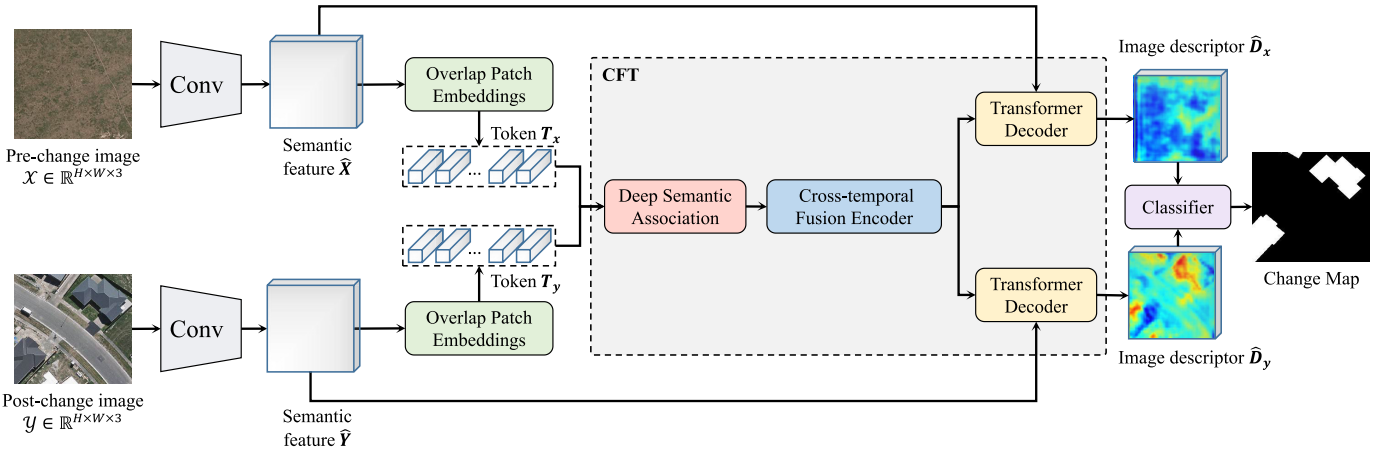


Fig. 2. Overview of the proposed CCLNet. For the input prechange and postchange images, CNN is first used to extract deep semantic features of the two images, respectively. Then, the deep features are converted into patches through overlap patch embeddings and sent to the CFT to fully model global context information. In CFT, the input tokens are first aggregated for DSA and then fed into the cross-temporal fusion encoder (CFE) for intratemporal and intertemporal long-range dependence modeling. The modeled tokens are sent to different TDs to restore them to pixel space. Finally, a classifier head is used to generate the final prediction using the bitemporal image descriptor.

information through dense connections. Besides, a designed integrated channel attention module (ECAM) is applied to refine the most representative features for classification. Due to the powerful advantages of transformers in long-range dependence modeling, some researchers began to introduce it in CD. Chen et al. [20] proposed a transformer-based CD method. The core idea of it is to convert the deep features extracted by CNN into a series of semantic tokens and send them to the transformer to mine global context information. Bandara and Patel [21] directly use the transformer encoder to extract deep features from the bitemporal image, and then use the MLP decoder to get the change prediction.

However, the above methods do not fully exploit the cross-temporal deep global context, resulting in insufficient global information critical for discovering changing objects. Therefore, we propose a CCLNet that exploits intra and intertemporal global modeling to learn the cross-temporal context for efficient CD in a concise manner.

B. Vision Transformer

The transformer network is originally proposed by Vaswani et al. [38] and used to process natural language processing (NLP) tasks. Completely different from CNN, it uses a multihead self-attention (MSA) mechanism to capture the long-range dependencies among different words, so it has a strong ability to extract global information. Since the transformer network shines in NLP, researchers consider applying it to the field of CV. Vision transformer (ViT) is first proposed for image classification by Dosovitskiy et al. [39]. Its core idea is to use position embedding to convert images into a set of tokens, and then use a transformer to encode these tokens. Since then, various transformer networks for CV tasks have sprung up. Carion et al. [40] applied ViT to the object detection task, which greatly simplified the object detection pipeline and realized end-to-end object detection. In order to achieve higher performance dense vision tasks (e.g., semantic segmentation), Wang et al. [41] integrated the idea of a feature pyramid into a transformer and proposed the pyramid ViT (PVT). Liu et al. [42] proposed a Swin transformer with an improvement in self-attention, where both the hierarchical structure and shifted window scheme are exploited. Yang et al. [43] proposed a lightweight transformer network with two enhanced self-attention mechanisms to improve model performance on mobile deployments. Although the above methods have achieved satisfactory performance in different applications, they cannot be directly applied to the CD task. In order to achieve a more efficient CD, we combine the advantages of transformer and CNN to design CCLNet.

III. Methodology

The overall framework of the proposed method is shown in Fig. 2. For the input bitemporal images, CNN is first applied for local modeling to obtain deep semantic features. Then, the overlapped patch embedding strategy that preserves spatial local continuity is designed to transform the semantic features into tokens. Next, a novel CFT is used to adequately model the cross-temporal global context. Finally, the features generated by CFT are fed to the prediction head to produce pixel-level predictions. We argue that the focus of context interaction should be in the encoding stage of deep semantic information, and context fusion across temporal is very necessary.

A. Feature Extraction

Existing studies have proved that CNN is suitable for early modeling of vision models [22]. Therefore, we first extract features from the input image using a CNN network. Here, considering the computational cost, ResNet-18 with some modifications is applied. In order to preserve more spatial details, we only use four stages of the network and set the stride of the last layer to 1, following a 3×3 convolution to reduce the number of channels to 32 to improve the subsequent computational efficiency. The generated features are again upsampled by a factor of two to reduce the loss of spatial detail. Let $\mathbf{X} \in \mathbb{R}^{3\times H\times W}$ and $\mathbf{Y} \in \mathbb{R}^{3\times H\times W}$ denote the prechange and postchange images, respectively, where the symbols H and W represent the height and width of the input

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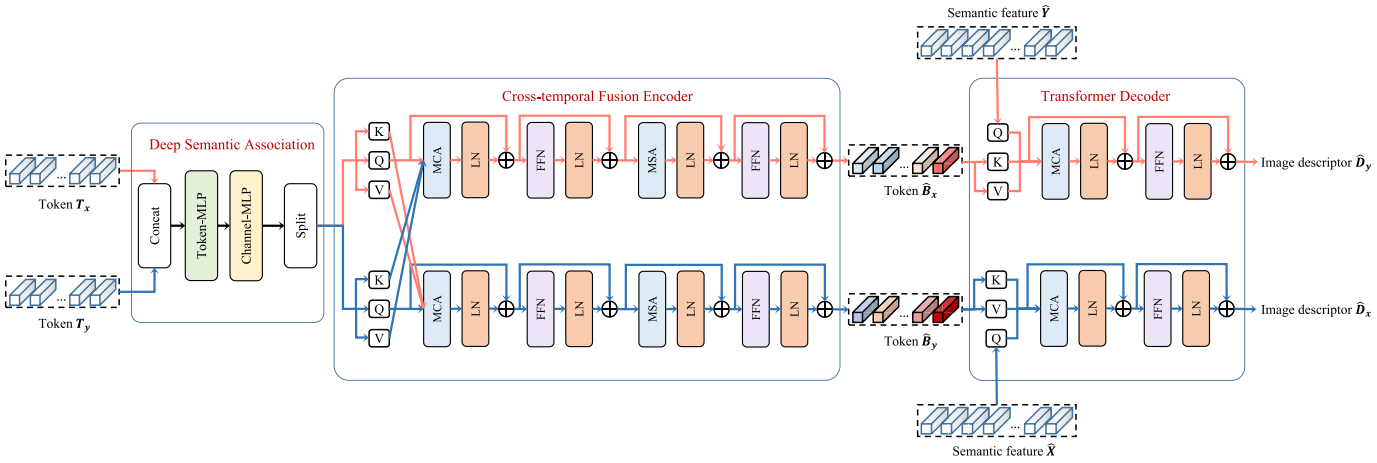


Fig. 3. Illustration of CFT. It consists of three units. The input bitemporal tokens are fed into the DSA unit together to first establish the spatial–temporal semantic association. Then, CFE models the intertemporal and intratemporal long-range dependencies of tokens to fully mine the cross-temporal global context information. Finally, the TD is used to restore the tokens to pixel space from a global perspective.

images. The extracted semantic features are represented as $\hat{\mathbf{X}} \in \mathbb{R}^{d\times h\times w}$ and $\hat{\mathbf{Y}} \in \mathbb{R}^{d\times h\times w}$ with $(w, h) = (W/4, H/4)$, where d represents the channel dimension.

To leverage CFT to model the global contextual information of semantic features, we follow ViT to divide the input semantic features $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ into a series of patches. In general, the original split method directly cuts the image into nonoverlapping patches, there is no continuity between each local patch, resulting in damage to local adjacent structures. We argue that overlapping split can mitigate this disadvantage. Let the patch size be K and the sliding window step size be S, the input features will be split into N patches, which can be formulated as

$$N = N_h \times N_w = \left[\frac{h-K+S}{S}\right] \times \left[\frac{w-K+S}{S}\right]. \quad (1)$$

Since there is an overlap region of size $(K-S)\times K$ between adjacent patches, the adjacent structural information is better preserved. In general, more overlapping areas lead to better performance, but at the same time increase the computational cost. Here, we use a convolution with kernel size $= 7$, stride $=$ 4, and padding $= 3$ for patch split. After splitting, the position embedding operation is used to transform patches into a series of tokens $\mathbf{T}_x \in \mathbb{R}^{n\times d}$ and $\mathbf{T}_y \in \mathbb{R}^{n\times d}$, $n = (w \times h)/16$.

B. Cross-Temporal Fusion Transformer

Existing studies have shown that the self-attention mechanism is preferred in modeling long-range dependencies and obtaining global context. Therefore, we design the CFT to fully explore the spatial–temporal long-range dependencies. As shown in Fig. 3, it is divided into three units, namely, DSA, CFE, and TD. The deep semantic association unit is responsible for fusing and encoding the deep features from two temporal images to establish the spatial–temporal association of cross-temporal semantics. Then, CFE exploits self-attention to fully explore the cross-temporal global semantic context. Finally, the TD is used to map the deep features into the pixel space.

1) Deep Semantic Association: For the bitemporal tokens $\mathbf{T}_x$ and $\mathbf{T}_y$, we first concatenate them together to obtain $\mathbf{T} \in \mathbb{R}^{2n\times d}$. Then, token-MLP and channel-MLP are used to model the long-range dependencies on spatial and channel it.

Without loss of generality, we rearrange the matrix $\mathbf{T} = [\mathbf{t}^1; \mathbf{t}^2; \ldots; \mathbf{t}^{2n}]$, where the vector $\mathbf{t}^i \in \mathbb{R}^d (1 \le i \le 2n)$. The token-MLP can be formulated as

$$\mathbf{h}^i = \mathbf{t}^i + \mathbf{W}_2 \mathrm{GELU}\left(\mathbf{W}_1 \mathrm{LN}\left(\mathbf{t}^i\right)\right), \quad i = 1, 2, \ldots, 2n \quad (2)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ denotes FC operation, GELU$(\cdot)$ represents activation function, and LN$(\cdot)$ represents LayerNorm. Thus, we obtain $\mathbf{H} = [\mathbf{h}^1; \mathbf{h}^2; \ldots; \mathbf{h}^{2n}]$, $\mathbf{h}^i \in \mathbb{R}^d (1 \le i \le 2n)$. We set $\mathbf{H}^{\mathrm{T}} = [\mathbf{h}_1; \mathbf{h}_2; \ldots; \mathbf{h}_d]$, $\mathbf{h}_i \in \mathbb{R}^{2n} (1 \le i \le d)$. The channel-MLP can be formulated as

$$\mathbf{b}_i = \mathbf{W}_4 \mathrm{GELU}(\mathbf{W}_3 \mathrm{LN}(\mathbf{h}_i)), \quad i = 1, 2, \ldots, d. \quad (3)$$

After (3), we get token $\mathbf{B}^{\mathrm{T}} = [\mathbf{b}_1; \mathbf{b}_2; \ldots; \mathbf{b}_d]$, $\mathbf{b}_i \in \mathbb{R}^{2n} (1 \le i \le d)$. $\mathbf{B}$ is split as a bitemporal token $\mathbf{B}_x \in \mathbb{R}^{n\times d}$ and $\mathbf{B}_y \in \mathbb{R}^{n\times d}$

2) Cross-Temporal Fusion Encoder: After establishing the spatial–temporal association of cross-temporal semantics, a CFE is used to further mine and aggregate intertemporal and intratemporal global information. First, multihead cross-attention (MCA) is used for intertemporal global information interaction. Given two tokens $\mathbf{B}_x$ and $\mathbf{B}_y$, the cross-temporal fusion process is formulated as

$$\begin{aligned}
\mathbf{Q}_1 &= \mathbf{B}_x \mathbf{W}_1^Q, \quad \mathbf{K}_1 = \mathbf{B}_x \mathbf{W}_1^K, \quad \mathbf{V}_1 = \mathbf{B}_x \mathbf{W}_1^V \\
\mathbf{Q}_2 &= \mathbf{B}_y \mathbf{W}_2^Q, \quad \mathbf{K}_2 = \mathbf{B}_y \mathbf{W}_2^K, \quad \mathbf{V}_2 = \mathbf{B}_y \mathbf{W}_2^V \\
\mathbf{Z}_1 &= \mathrm{LN}(\mathrm{MCA}(\mathbf{Q}_1, \mathbf{K}_2, \mathbf{V}_2)) + \mathbf{Q}_1 \\
\tilde{\mathbf{B}}_x &= \mathrm{LN}(\mathrm{FFN}(\mathbf{Z}_1)) + \mathbf{Z}_1 \\
\mathbf{Z}_2 &= \mathrm{LN}(\mathrm{MCA}(\mathbf{Q}_2, \mathbf{K}_1, \mathbf{V}_1)) + \mathbf{Q}_2 \\
\tilde{\mathbf{B}}_y &= \mathrm{LN}(\mathrm{FFN}(\mathbf{Z}_2)) + \mathbf{Z}_2
\end{aligned} \quad (4)$$

where $\mathbf{W}_i^Q$, $\mathbf{W}_i^K$, and $\mathbf{W}_i^V$ denote different learnable weight matrices. The feed-forward network (FFN) composed of the MLP layer and GELU activation layer performs channel modeling on the features produced by MCA and the layer normalization (LN) layer is applied after MCA and FFN. The

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attention score of MCA is jointly generated by $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, which can be formulated as

$$\mathbf{Z} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \tag{5}$$

where d is the dimension of the query and key. As shown in (4), $\mathbf{Q}_1$, $\mathbf{K}_2$, and $\mathbf{V}_2$ from different temporal tokens calculate attention scores. Through this computation, information from different times can be fully fused and interacted, focusing on the discriminative features of changing regions. After intertemporal information fusion, we get $\tilde{\mathbf{B}}_x \in \mathbb{R}^{n\times d}$ and $\tilde{\mathbf{B}}_y \in \mathbb{R}^{n\times d}$. Then, we use self-attention to effectively integrate the global interaction of the intratemporal information. Take input token $\tilde{\mathbf{B}}_x \in \mathbb{R}^{n\times d}$ as an example, the process of information fusion within the temporal can be formulated as

$$\begin{aligned}\mathbf{Q} &= \tilde{\mathbf{B}}_x\mathbf{W}^Q, \quad \mathbf{K} = \tilde{\mathbf{B}}_x\mathbf{W}^K, \ \mathbf{V} = \tilde{\mathbf{B}}_x\mathbf{W}^V \\ \mathbf{Z} &= \text{LN}(\text{MSA}(\mathbf{Q},\mathbf{K},\mathbf{V})) + \mathbf{Q} \\ \hat{\mathbf{B}}_x &= \text{LN}(\text{FFN}(\mathbf{Z})) + \mathbf{Z}.\end{aligned} \tag{6}$$

Here, we use MSA instead of self-attention, because MSA can model from multiple subfeature spaces and obtain information from multiple perspectives. In our work, it is set to 64 attention heads. The cascaded MCA and MSA alternately fuse information within intertemporal and intratemporal.

3) Transformer Decoder: After fully exploring the global contextual relations in intratemporal and intertemporal, we obtain two sets of semantic tokens $\hat{\mathbf{B}}_x$ and $\hat{\mathbf{B}}_y$ rich in discriminative information. In order to further learn the cross-temporal context in the decoder, $\hat{\mathbf{B}}_x$ and $\hat{\mathbf{B}}_y$ are decoded with the semantic features at different times to obtain the final image descriptor, as shown in Fig. 3. Taking token $\hat{\mathbf{B}}_y \in \mathbb{R}^{n\times d}$ as an example, TD explores the relationship between each pixel in the semantic feature $\hat{\mathbf{X}}$ and the token $\hat{\mathbf{B}}_y$ to obtain a refined image descriptor $\hat{\mathbf{D}}_x \in \mathbb{R}^{d\times h\times w}$. Specifically, the pixels in $\hat{\mathbf{X}}$ are regarded as queries, and the token is used as key and value. The decoding process can be formulated as

$$\begin{aligned}\mathbf{Q} &= \hat{\mathbf{X}}\mathbf{W}_1^Q, \quad \mathbf{K} = \hat{\mathbf{B}}_y\mathbf{W}_1^K, \quad \mathbf{V}_1 = \hat{\mathbf{B}}_y\mathbf{W}_1^V \\ \mathbf{D}_x &= \text{LN}(\text{MCA}(\mathbf{Q},\mathbf{K},\mathbf{V})) + \mathbf{Q} \\ \hat{\mathbf{D}}_x &= \text{LN}(\text{FFN}(\mathbf{D}_x)) + \mathbf{D}_x.\end{aligned} \tag{7}$$

C. Prediction Head

After getting the final image descriptor $\hat{\mathbf{D}}_x$ and $\hat{\mathbf{D}}_y$, a prediction head with shallow FCN is used for generating change probability maps $P \in \mathbb{R}^{H\times W\times 2}$, which can be formulated as follows:

$$P = \text{softmax}\left(f\left(g\left(\left|\hat{D}_x - \hat{D}_y\right|\right)\right)\right) \tag{8}$$

where $g(\cdot)$ denotes the upsampling quadruple operation, $f(\cdot)$ is the shallow FCN, and softmax$(\cdot)$ indicates softmax function. Specifically, the shallow FCN consists of two convolutional layers. The convolution of the first layer is followed by BatchNorm and ReLU operation. The convolution of the second layer reduces the number of feature channels to 2. The kernel size of both convolutions is 3×3. The prediction mask $M \in \mathbb{R}^{H\times W}$ is calculated by the pixel-level argmax operation on the channels of P.

During training, the cross-entropy loss function is used to optimize the model parameters, which can be formulated as

$$L_{\text{ce}} = -\sum_{i=1}^{H*W} y_i \log\left(\hat{y}_i\right) \tag{9}$$

where y_i represents the ground truth of the ith pixel, and $\hat{y}_i$ represents the predicted probability of the ith pixel.

D. CCLNet++

Although CCLNet has achieved good performance at a relatively small computing cost, considering that in some applications, the power of the computing platform is sufficient, and performance is the priority at this time. Therefore, a performance version of CCLNet++ is designed to achieve better detection. In CCLNet++, we replace ResNet-18 with Res2Net-50 [44]. The last two stages of the network made the same modifications as ResNet-18. The reason for using Res2Net as the backbone in CCLNet++ is that it enhances the multiscale representation ability of features, which can improve the detection performance of the method for multiscale objects.

IV. Experiments

In this section, mainstream CD datasets and evaluation protocols are first introduced. Then, we elaborate on the implementation details. Next, extensive experimental results are analyzed to demonstrate the effectiveness of the proposed network. Finally, the ablation experiments are described.

A. Experimental Setup

1) Implementation Details: ResNet-18 pretrained on ImageNet [45] is used as our backbone to extract features. The model is implemented on pytorch [46] and trained on a single RTX3090. During training, the batchsize is 8, the initial learning rate is 0.01 and decays linearly to 0, and the epoch is 200. Flip, rescale, crop, and Gaussian blur are applied to the data augmentation of the input image. All input images are resized to 256×256. The optimizer uses stochastic gradient descent (SGD) with momentum set to 0.9 and weight decay to 0.0005. After each round of epoch, the model is verified on the verification set, and the model with the best verification effect is used to evaluate the test set.

2) Datasets: Experiments are performed on three mainstream CD datasets, which are described in detail as follows.

LEVIR-CD [12] is a large building CD dataset with 637 pairs of high-resolution images of size 1024×1024. In the same way as [21], we slice it into nonoverlapping 256×256 patches and divide the dataset into train/validation/test of quantity 7120/1024/2048.

WHU-CD [24] is a dataset for building CD, which in total contains a pair of high-resolution remote sensing images with a resolution of $32\,507 \times 15\,345$. Similar to [20], we crop the image nonoverlapping into 256×256 resolution patches. The cropped patches are randomly divided into three parts for training, validation, and testing: 6096/762/762.

DSIFN-CD [23] is a dataset that contains changes in different land cover objects. Its satellite imagery comes from six

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TABLE I
COMPARISON WITH THE NINE STATE-OF-THE-ART METHODS ON DIFFERENT DATASETS. "*" MEANS METHOD USING TRANSFORMER AS BACKBONE. RED AND GREEN DENOTE THE BEST AND THE SECOND BEST RESULTS

Method	LEVIR-CD [12]					WHU-CD [24]					DSIFN-CD [23]				
	Precision	Recall	F1	IoU	OA	Precision	Recall	F1	IoU	OA	Precision	Recall	F1	IoU	OA
FC-EF [27]	86.91	80.17	83.40	71.53	98.39	71.63	67.25	69.37	53.11	97.61	72.61	52.73	61.09	43.98	88.59
FC-Siam-Di [27]	89.53	83.31	86.31	75.92	98.67	47.33	77.66	58.81	41.66	95.63	59.67	65.71	62.54	45.50	86.63
FC-Siam-Conc [27]	91.99	76.77	83.69	71.96	98.49	60.88	73.58	66.63	49.95	97.04	66.45	54.21	59.71	42.56	87.57
DTCDSCN [15]	88.53	86.83	87.67	78.05	98.77	63.92	82.30	71.95	56.19	97.42	53.87	77.99	63.72	46.76	84.91
STANet [12]	83.81	**91.00**	87.26	77.40	98.66	79.37	85.50	82.32	69.95	98.52	67.71	61.68	64.56	47.66	88.49
IFNet [23]	**94.02**	82.93	88.13	78.77	98.87	**96.91**	73.19	83.40	71.52	98.83	67.86	53.94	60.10	42.96	87.83
SNUNet [37]	89.18	87.17	88.16	78.83	98.82	85.60	81.49	83.50	71.67	98.71	60.60	72.89	66.18	49.45	87.34
BIT [20]	89.24	89.37	89.31	80.68	98.92	86.64	81.48	83.98	72.39	98.75	68.36	70.18	69.26	52.97	89.41
ChangeFormer* [21]	92.05	88.80	90.40	82.48	**99.04**	**90.59**	**89.65**	**90.11**	**82.01**	**99.18**	**88.48**	**84.94**	**86.67**	**76.48**	**95.56**
CCLNet	91.15	89.85	**90.49**	**82.64**	**99.04**	86.92	89.37	88.13	78.78	99.00	85.16	76.85	80.79	67.78	93.79
CCLNet++	**92.31**	**89.88**	**91.08**	**83.62**	**99.10**	89.83	**90.78**	**90.31**	**82.32**	**99.19**	**90.96**	**86.11**	**88.47**	**79.32**	**96.19**

cities in China. We crop all images to 256 × 256 nonoverlapping patches and follow the author's default training, validation, and testing sets, resulting in 14 400/1360/192 samples for training, testing, and validation.

3) Evaluation Protocol: In the experiments, we use five indicators commonly used in CD to compare with the SOTA method, such as F1-score, precision, recall, intersection over union (IoU), and overall accuracy (OA). If the effect of the method is better, the values of these indicators are larger. Among them, F1-score and IoU are the main quantitative indicators. All indicators are defined as follows:

$$\begin{aligned} \text{precision} &= \text{TP}/(\text{TP}+\text{FP}) \\ \text{recall} &= \text{TP}/(\text{TP}+\text{FN}) \\ \text{F1} &= \frac{2}{\text{precision}^{-1}+\text{recall}^{-1}} \\ \text{IoU} &= \text{TP}/(\text{TP}+\text{FN}+\text{FP}) \\ \text{OA} &= (\text{TP}+\text{FN})/(\text{TP}+\text{TN}+\text{FN}+\text{FP}) \end{aligned} \tag{10}$$

where TP, TN, FP, and FN are true positive, true negative, false positive, and false negative, respectively.

B. Comparison to State-of-the-Art

We compare our method with existing state-of-the-art (SOTA) CD methods, including CNN-based methods and transformer-based methods. CNN-based methods include fully convolutional early fusion (FC-EF) [27], FC-Siam-Di [27], FC-Siam-Conc [27], DTCDSCN [15], STANet [12], IFNet [23], and SNUNet [37]. Transformer-based methods include bitemporal image transformer (BIT) [20] and ChangeFormer [21]. The results of all methods are generated either on publicly available data or by retraining the model with code released by the authors. Each method is described in detail as follows.

1) FC-EF is an image-level CD method. The input bitemporal images are concatenated and directly sent to an FCN to obtain the detection results.
2) FC-Siam-Di is a feature-level fusion method that uses a Siamese FCN to extract multilevel features, and then uses feature differences to fuse information to generate change predictions.
3) FC-Siam-Conc is a feature-level fusion method. Similar to FC-Siam-Di, a Siamese FCN first extracts multilevel features. Then, bitemporal features are concatenated to generate change predictions.
4) DTCDSCN adds channel attention and spatial attention to deep Siamese FCN. In addition, semantic segmentation decoders belonging to bitemporal images are used to improve the discriminative ability of features for different objects.
5) STANet is a metric-based Siamese FCN approach that utilizes spatial–temporal attention to obtain discriminative features.
6) IFNet uses the designed attention module to fuse features of different scales in the decoding process to obtain change predictions.
7) SNUNet is a method based on multiscale feature fusion. Dense connections are applied to Siamese networks to generate change predictions.
8) BIT first uses the convolutional network to extract features. Next, the transformer encoder and decoder are used to model the context of space and time to obtain detection results.
9) ChangeFormer uses the transformer encoder as the backbone to extract features, and then uses the MLP decoder to generate change predictions.

1) Performance on LEVIR-CD: Table I shows the results of our method compared with the SOTA methods. It can be observed that CCLNet outperforms other SOTA methods on two main metrics F1 and IoU, especially compared with ChangeFormer which uses a hierarchically structured transformer as the backbone. In the performance version of CCLNet++, our results go even further. This demonstrates that our method is able to fully explore the global context across temporal and achieve better change predictions.

The visualization of the detection results of different methods is shown in Fig. 4. It can be observed that our method can more accurately locate and segment the objects. In the first and seventh rows, only our method accurately localizes

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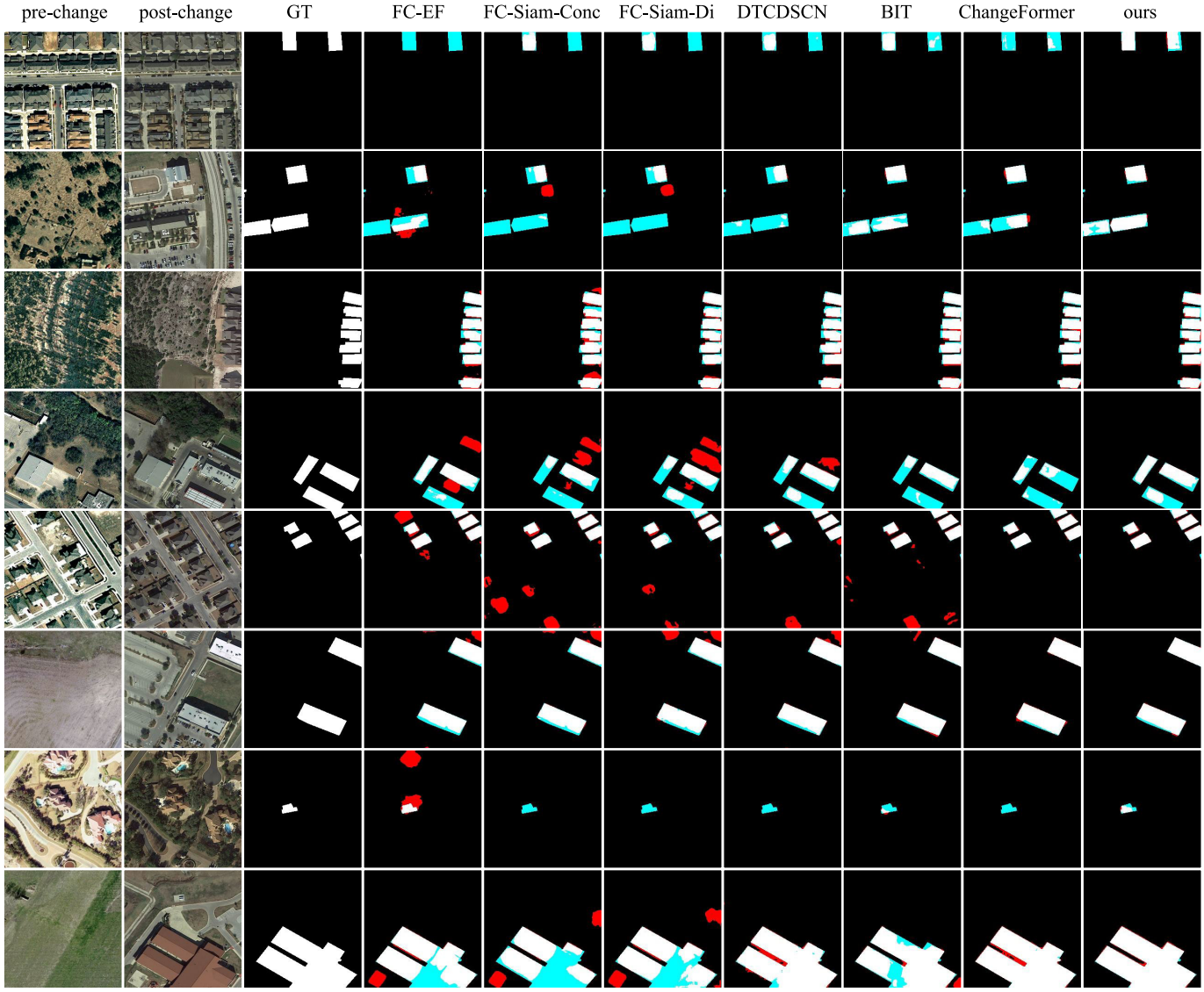


Fig. 4. Visual comparison of different methods on LEVIR-CD dataset. The red part represents FP, the blue part represents FN, the black part represents TN, and the white part represents TP.

the changing objects in the cluttered background. Among other rows, our method makes fewer FP predictions and produces more complete predictions than other methods. The visualization results illustrate that our method makes full use of cross-temporal global context to achieve superior results.

2) Performance on WHU-CD: The comparison results of CCLNet and other SOTA methods are shown in Table I. All metrics of CCLNet outperform CNN backbone methods. Most of the indicators of CCLNet++ are better than all SOTA methods. Compared with ChangeFormer with the transformer as the backbone, the recall is improved from 89.65 to 90.78, the F1 is improved from 90.11 to 90.31, and the IoU is improved from 82.01 to 82.32. We attribute this improvement to the powerful cross-temporal long-range dependence modeling capabilities of the CFT.

3) Performance on DSIFN-CD: Table I shows the comparison of CCLNet and other SOTA methods on the DSIFN-CD dataset. Among the methods that also use CNN as a backbone, CCLNet achieves the best results. CCLNet++ surpasses all SOTA methods, including the ChangeFormer. The results show that CCLNet achieves superior performance even with a naive backbone network. After replacing the more powerful CNN backbone, our method can even surpass the method using a transformer as a backbone, which is due to the powerful cross-temporal context modeling ability of the CFT.

TABLE II

COMPARISON WITH THE MODEL SIZE AND AVERAGE TEST SPEED OF THREE STATE-OF-THE-ART MODELS. THE INFERENCE SPEED OF EACH METHOD IS CALCULATED WITH 256 × 256 IMAGES. RED AND GREEN DENOTE THE BEST AND THE SECOND BEST RESULTS

	DTCDSCN	BIT	ChangeFormer	CCLNet	CCLNet++
Params (M)	41.07	**11.94**	41.02	**14.25**	28.78
MACs (G)	13.18	**8.13**	197.16	**8.16**	23.27
FPS	33.09	**38.91**	16.56	**57.85**	25.28
Time (ms)	30.22	**25.70**	60.38	**17.28**	39.55

C. Computational Complexity Analysis

To further compare the efficiency of different methods, we evaluate the complexity of different models on an RTX3090. As shown in Table II, evaluation metrics include

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TABLE III

ABLATION EXPERIMENTS OF CFT ON TWO DATASETS. DSA STANDS FOR DEEP SEMANTIC ASSOCIATION, CFE DENOTES CROSS-TEMPORAL FUSION ENCODER, AND TD MEANS TRANSFORMER DECODER

No.	Component			LEVIR-CD [12]			DSIFN-CD [23]		
	DSA	CFE	TD	F1	IoU	OA	F1	IoU	OA
#1	-	-	-	87.50	77.78	98.78	61.39	44.29	89.48
#2	-	✓	✓	88.21	78.91	98.86	66.36	49.66	89.77
#3	✓	-	✓	88.64	79.61	98.89	65.68	48.89	90.38
#4	✓	✓	-	88.01	78.59	98.78	70.46	54.40	89.32
#ours	✓	✓	✓	**90.49**	**82.42**	**99.03**	**80.79**	**67.78**	**93.79**

training parameters (Params), multiply-accumulate operations (MACs), frames per second (FPS), and inference time per image (time). It can be observed from the results that CCLNet outperforms DTCDSCN and ChangeFormer utilizing multilevel feature fusion in all metrics. Compared with BIT, which also uses a lightweight backbone, CCLNet is almost equivalent to but slightly inferior to BIT in terms of Params and MACs but surpasses it in FPS and time. Although the indicators of CCLNet++ are not dominant, compared with ChangeFormer, it is not only superior to it in detection performance but also in network complexity. The experimental results show that CCLNet can achieve satisfactory performance in both detection performance and inference speed. After replacing the more powerful backbone, CCLNet++ can still achieve a better tradeoff between performance and complexity. We attribute this to two aspects. First, CCLNet directly models the deepest semantic features, avoiding the huge computational overhead of multilevel feature fusion. Second, the powerful intratemporal and intertemporal global modeling capabilities of the CFT module make features more discriminative so as to improve detection performance.

D. Ablation Studies

1) Effectiveness of CFT: To verify the effectiveness of CFT, we removed it from the CCLNet for experiments and conducted experiments on LEVIR-CD and DSIFN-CD datasets, the results are shown in Table III. No. #1 represents the network with CFT removed. It can be observed that after removing CFT, all indicators have declined, especially in the DSIFN-CD dataset. The experimental results show that CFT contributes a lot to the improvement of CD performance, which verifies the importance of cross-temporal context learning.

To further study the contribution of different components in CFT to the performance, ablation experiments of the three components are also carried out. No. #2, No. #3, and No. #4 represent experiments after removing DSA CFE, and TD, respectively. Experimental results show that all three components contribute to the performance improvement. On the LEVIR dataset, the performance gap is not big after removing each component. On the large dataset DSIFN, removing CFE and DSA makes the performance drop a lot, which shows that these two components are the key components in CFT, especially CFE. This shows that by modeling intertemporal and intratemporal long-distance dependencies, the learning ability of cross-temporal context can be greatly improved, thus obtaining satisfactory performance.

TABLE IV

ABLATION EXPERIMENTS OF CFE LAYERS ON TWO DATASETS

Depth	LEVIR-CD [12]			DSIFN-CD [23]		
	F1	IoU	OA	F1	IoU	OA
1	**90.49**	**82.42**	99.03	**80.79**	**67.78**	**93.79**
2	90.34	82.39	99.02	78.30	64.35	93.05
3	90.44	82.56	**99.04**	76.04	61.34	91.65

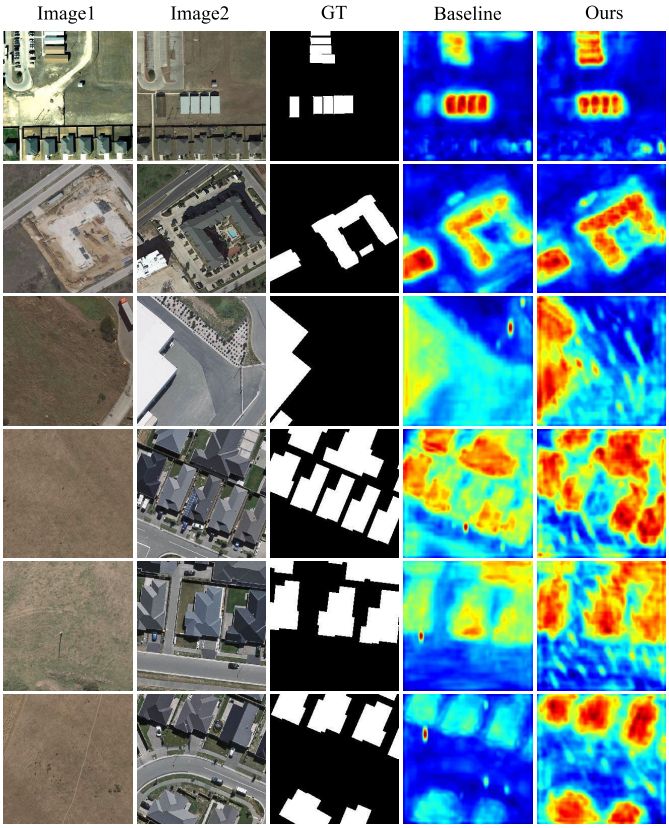


Fig. 5. Visual comparison of features maps $|\hat{D}_x - \hat{D}_y|$ generated by baseline and CFT. Redder colors indicate higher attention scores.

2) Layers of Cross-Temporal Fusion Encoder in CFT: The depth of the transformer encoder is a crucial factor affecting the performance and efficiency of the transformer. Hence, we set up different numbers of CFE layers for experiments, as shown in Table IV. After increasing the number of CFE layers, the performance gap on the LEVIR-CD is not large, but there is a very obvious difference on the DSIFN-CD.

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As the number of CFE layers increases, the performance of DSFIN-CD gradually decreases. We argue that too many CFE layers make it difficult for the network to converge, leading to this result.

3) Qualitative Result of CFT: To further demonstrate the superiority of CFT, we visualize the feature maps $|\hat{D}_x - \hat{D}_y|$ generated by CFT and compare them with the baseline that CFT is removed. As shown in Fig. 5, after encoding by CFT, the change object in the feature map becomes more significant, and the key parts of the object are paid more attention to. The visualization results show that CFT enhances the discriminative features of objects through cross-temporal context learning.

V. Conclusion

Finding changing objects of interest is a challenging task due to the complexity and variety of objects in a changing scene. Most existing methods fail to adequately explore the cross-temporal global context, making the establishment of spatial–temporal deep global associations insufficient. In this article, we propose a cross-temporal context learning network, called CCLNet, for efficient CD of remote sensing images. The core idea of CCLNet is to realize full cross-temporal context exploration through intertemporal and intratemporal long-range dependence mining. On one hand, the feature discriminativeness of change objects is improved by fully exploring the cross-temporal context. On the other hand, this efficient global context representation enables us to achieve reliable performance with a lightweight backbone, thereby reducing computational costs. Specifically, a lightweight CNN is first used to extract deep semantic features. Then, a well-designed CFT is proposed to establish long-range dependencies between deep semantics to find changing objects in the scene. Experiments on three benchmarks show that our proposed method can perform the CD task faster than SOTA methods while maintaining comparable matching accuracy.

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